Silicon Photonic Test Structures for TE and TM Strip Waveguides: Mach-Zehnder Interferometers, Ring Resonators, Bragg Gratings and a Directional Coupler

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***Abstract* —** *This report describes unbalanced Mach-Zehnder interferometers (MZIs) to be built from 500 nm × 220 nm silicon strip waveguides, for both the quasi-TE and quasi-TM modes. The TE mode uses path-length differences of 50, 100 and 200 µm, and the designed TM device (MZI4) has ΔL = 49.494 µm (approximately 49.5 µm); the simulation study also covers 100 and 200 µm for TM. Two all-pass ring resonators are added for the TE mode as an independent check of the group index. The goal is to extract the waveguide group index from the measured free spectral range (FSR) and compare it with mode-solver predictions. The report covers the waveguide model, the interferometer and ring transfer functions, the derivation of the group-index extraction equation, circuit simulations, a corner analysis of width and thickness variation, and the planned measurement analysis. The corner analysis predicts the variation in effective index, group index, FSR and MZI fringe and ring-resonance wavelengths. Additional Bragg-grating wavelength shifts are estimated separately. The submitted layout also contains a TE and a TM Bragg grating, a directional coupler and a reference loopback.*

I. Introduction

The Mach-Zehnder interferometer is a basic building block of silicon photonics. It underlies modulators, switches, wavelength filters and sensors [1-3]. In all of these, device performance depends directly on how accurately we know the waveguide group index, n_g. That index sets the FSR of filters, the bandwidth of modulators, and the delay in true-time-delay lines [4].

An unbalanced MZI gives a simple, direct way to measure n_g. The FSR of the output spectrum depends only on the path-length difference ΔL and the group index. Because ΔL is defined lithographically and known precisely, a measured FSR maps directly to n_g [4].

The objectives of this design are:

- Extract n_g for the fundamental quasi-TE and quasi-TM modes of a 500 nm × 220 nm strip waveguide from measured MZI spectra.
- Verify that the FSR scales as 1/ΔL by using three imbalance lengths for the TE mode.
- Extract n_g independently from two TE ring resonators and observe how the coupling gap changes resonance linewidth and extinction ratio.
- Compare the extracted n_g with simulation and attribute any differences to fabrication variation in waveguide width and thickness. The expected size of these differences is set by the corner analysis in Section III-H.
- Test other building blocks of the platform on the same chip: a TE Bragg grating and a directional coupler, simulated here and to be measured, a TM Bragg grating whose stopband is only estimated, and a loopback that gives the grating coupler response.

II. Theory

A. MZI Transfer Function

A Y-branch splits the input field E_i equally into two arms, so each arm carries $E_i/\sqrt{2}$. After propagating through arms of length L_1 and L_2, with propagation constants β_1 and β_2 and power attenuation α, the fields reaching the combiner are:

$$E_{1,2} = \frac{E_i}{\sqrt{2}} e^{-j\beta_{1,2}L_{1,2} - \alpha L_{1,2}/2} \tag{1}$$

The Y-branch combiner sends half of each input field into the fundamental output mode; the rest is lost to radiation modes [4], [9]. The output field is therefore:

$$E_o = \frac{E_1 + E_2}{\sqrt{2}} = \frac{E_i}{2}\left(e^{-j\beta_1 L_1 - \alpha L_1/2} + e^{-j\beta_2 L_2 - \alpha L_2/2}\right) \tag{2}$$

Both arms use the same waveguide cross-section, so $\beta_1 = \beta_2 = \beta = 2\pi n_{eff}/\lambda$. Neglecting loss, the intensity transfer function is:

$$T_{MZI}(\lambda) = \frac{I_o}{I_i} = \frac{1}{2}[1 + cos(\beta \Delta L)], \;\; \Delta L = L_2 - L_1 \tag{3}$$

With loss included, the arms have slightly different amplitudes, which limits the extinction ratio. At a propagation loss of 3 dB/cm, a 200 μm path-length difference produces a 0.06 dB difference in power attenuation. The corresponding field-amplitude ratio is approximately 0.993, meaning that the more attenuated arm has about 0.7% lower field amplitude. This small imbalance suggests that deep nulls should remain visible in the simulated spectra.

B. Group Index from the Free Spectral Range

The phase difference between the two arms is:

$$\varphi(\lambda) = \frac{2\pi n_{eff}(\lambda)\Delta L}{\lambda} \tag{4}$$

Adjacent transmission maxima occur at wavelengths λ_m and λ_{m+1}, where the phase changes by exactly 2π. The FSR is small compared with λ, so we expand φ to first order about the midpoint wavelength λ:

$$|\Delta\varphi| \approx \left|\frac{d\varphi}{d\lambda}\right| \cdot FSR = 2\pi \tag{5}$$

Differentiating (4), and remembering that n_{eff} itself depends on λ, gives

$$\frac{d\varphi}{d\lambda} = -\frac{2\pi\Delta L}{\lambda^2}\left(n_{eff} - \lambda\frac{dn_{eff}}{d\lambda}\right) = -\frac{2\pi\Delta L n_g}{\lambda^2} \tag{6}$$

The bracketed term is the definition of the group index, $n_g = n_{eff} - \lambda \cdot dn_{eff}/d\lambda$ [4]. Substituting (6) into (5):

$$FSR = \frac{\lambda^2}{n_g \Delta L} \quad \Longrightarrow \quad n_g(\lambda) = \frac{\lambda^2}{\Delta L \cdot FSR(\lambda)} \tag{7}$$

Equation (7) is the extraction equation. The FSR is set by n_g, not n_{eff}, because waveguide dispersion shifts the fringe spacing. Each measured fringe spacing gives one value of n_g, assigned to the midpoint wavelength of that fringe pair. A single spectrum therefore yields $n_g(\lambda)$ across the measurement band.

<u>Test with simulation data:</u> From the INTERCONNECT spectrum of MZI1 (TE, ΔL = 50 μm), adjacent minima occur near 1543.51 and 1554.92 nm. Using the unrounded minima positions, the FSR is 11.416 nm at λ = 1549.215 nm. Using (7), the extracted group index is $n_g = \lambda^2 / (\Delta L \cdot FSR) =$

(1.549215 μm)² / (50 μm × 11.416 nm) = 4.205, which agrees with the MODE value of 4.204 to within 0.03%. Applying the same procedure to all adjacent pairs of minima in this spectrum gives group-index values within 0.05% of the MODE prediction across 1510 to 1590 nm.

C. Ring Resonator

An all-pass ring resonator consists of a ring of radius R side-coupled to a straight bus waveguide. With self-coupling coefficient r at the coupler and single-pass amplitude transmission a, the through-port transmission is [7]:

$$T_{ring}(\lambda) = \frac{a^2 - 2ra\cos\varphi + r^2}{1 - 2ra\cos\varphi + (ra)^2}, \quad \varphi = \beta L, \quad L = ring\ circumference \tag{8}$$

Resonances occur when the round-trip phase φ is a multiple of 2π, producing dips in the through-port spectrum. The derivation in Section II-B applies directly with ΔL replaced by the ring round-trip length L:

$$FSR = \frac{\lambda^2}{n_g L} \quad \Longrightarrow \quad n_g = \frac{\lambda^2}{L \cdot FSR} \tag{9}$$

The coupling gap sets r. A smaller gap gives stronger coupling, lower r and wider resonances. When r equals a (critical coupling), the resonance dips reach their maximum depth [7], [8]. Varying the gap at fixed radius therefore changes linewidth and extinction ratio but leaves the FSR unchanged, which isolates the effect of coupling.

III. Modelling and Simulation

A. Waveguide Geometry

The waveguide is a silicon strip, 500 nm wide and 220 nm thick, on a buried oxide layer with SiO_2 top cladding, the standard platform for silicon photonic wires [5], [11]. The fundamental quasi-TE and quasi-TM modes are both studied. The 500 nm waveguide width matches the standard grating-coupler and Y-branch library components [9–11], allowing the devices to be connected without additional transition tapers. The supported modes and their effective indices were evaluated using the Lumerical MODE eigenmode solver.

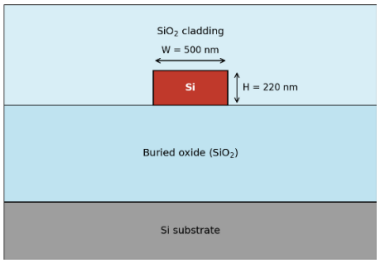


Fig. 1. Cross-section of the silicon strip waveguide.

B. Mode Profiles

The modes were computed with Lumerical MODE Solutions using the finite-difference eigenmode solver [6], [13] at λ = 1550 nm. Figs. 2 and 3 show the electric-field intensity of the quasi-TE and quasi-TM modes. The quasi-TE mode is predominantly confined within the silicon core. The quasi-TM mode exhibits a different spatial field distribution, with greater extension into the surrounding cladding. These differences in modal confinement influence the sensitivity of each polarization to waveguide geometry and fabrication variations.

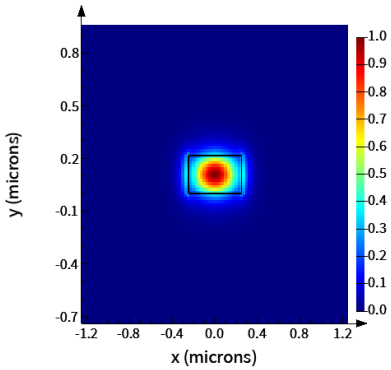


Fig. 2. Electric-field intensity of the quasi-TE mode at 1550 nm.

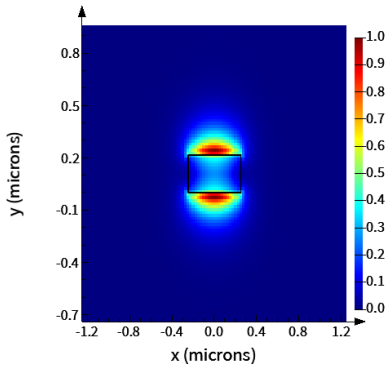


Fig. 3. Electric-field intensity of the quasi-TM mode at 1550 nm.

C. Effective and Group Index

A wavelength sweep from 1500 to 1600 nm, including material dispersion in Si and SiO_2, gives the effective and group indices shown in Figs. 4 and 5. Table I summarizes the values at 1550 nm. The simulated effective index decreases with wavelength over this band, consistent with wavelength-dependent modal confinement and material dispersion. The group index differs substantially from the effective index because it also depends on the wavelength derivative of the effective index. This distinction is important when predicting interferometer fringe spacing and ring-resonance spacing.

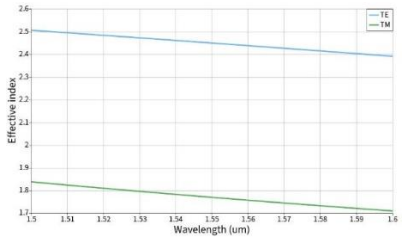


Fig. 4. Effective index versus wavelength for the quasi-TE and quasi-TM modes.

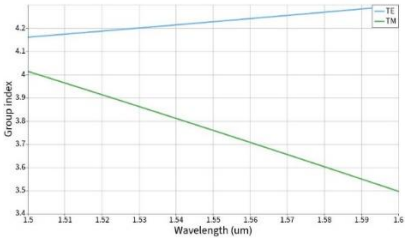


Fig. 5. Group index versus wavelength for the quasi-TE and quasi-TM modes.

TABLE I. SIMULATED MODE PROPERTIES AT 1550 NM

Mode	n_{eff}	n_g
Quasi-TE	2.447	4.204
Quasi-TM	1.769	3.720

D. Compact Model

For circuit simulation and data fitting, the effective index is described by a second-order Taylor expansion about $\lambda_0 = 1.55$ μm [4]:

$$n_{eff}(\lambda) = n_1 + n_2(\lambda - \lambda_0) + n_3(\lambda - \lambda_0)^2 \tag{10}$$

Applying the group-index definition to (10) at $\lambda = \lambda_0$ gives a direct link between the compact model and the group index:

$$n_g(\lambda_0) = n_1 - \lambda_0 n_2 \tag{11}$$

The fitted coefficients (with λ in μm) are listed in Table II. Using the fitted coefficients, (11) gives ng = 4.204 for TE and ng = 3.720 for TM, consistent with the values in Table I to within 0.002. The coefficients were obtained by fitting the MODE effective-index data over 1500 to 1600 nm, with a maximum residual below 4×10^{-5}.

TABLE II. COMPACT MODEL COEFFICIENTS ($\lambda_0 = 1.55$ μm)

Mode	n_1	n_2 (μm^{-1})	n_3 (μm^{-2})
Quasi-TE	2.4468	−1.1334	−0.0439
Quasi-TM	1.7688	−1.2587	1.9151

E. Design Variations

Table III lists the six interferometers and rings of the layout. The ΔL values were chosen so that every design fits several fringes in the ~50 nm measurement bandwidth. The smallest imbalance, 50 μm, gives about four fringes. The largest imbalance, 200 μm, provides approximately 17 fringe spacings across a 50 nm band, before accounting for the usable spectral envelope. Expected FSRs were calculated from (7) at 1550 nm, using ng = 4.204 for TE and 3.720 for TM from Table I. Ten circuits are in the submitted layout: the TE interferometers MZI1, MZI2 and MZI3, the TM interferometer MZI4, the rings Ring1 and Ring2, and four additional structures described in Section IV (Bragg1, BraggTM1, DC1 and Loopback1). MZI5 and MZI6 (TM, ΔL = 100 and 200 μm) are kept in the simulation study but were left out of the chip, because the 605 μm × 410 μm area has room for only ten circuits with two grating couplers each. The layout of MZI4 specifies arm lengths of 200.001 μm and 249.495 μm, giving ΔL = 49.494 μm, approximately 49.5 μm.

TABLE III. DESIGN VARIATIONS AND EXPECTED FSR AT 1550 NM

Device	Polarization	ΔL or ring (μm)	Expected FSR (nm)
MZI1	TE	50	11.43
MZI2	TE	100	5.72
MZI3	TE	200	2.86
MZI4	TM	49.5	13.05
Ring1	TE	R = 10, gap 0.20	8.37
Ring2	TE	R = 10, gap 0.30	8.37

The FSR also follows the expected 1/ΔL trend: halving ΔL doubles the FSR. The round-trip length of each ring is approximately L = πR + 36.9 μm = 68.3 μm. This comprises the semicircular bend section with R=10 μm and the connecting waveguide section. The round-trip length, not the circumference of a full circle, is used to calculate the expected FSR. This gives about six resonances in the measurement band. The two rings share the same radius, so their FSRs should match; only the gap differs. The MZI, ring and directional-coupler access waveguides use nominal 500 nm strip widths. The Bragg gratings use the width corrugations specified in Table VII. The nominal straight-waveguide cross-section is 500 nm × 220 nm, with fabrication variations considered in Section III-H. TE devices use 10 μm bend radii. TM devices use larger bend radii, because the TM mode is less confined and has higher bend loss at small radii [4].

F. Circuit Simulation

The circuits were modelled in Lumerical INTERCONNECT [13] (Fig. 6). Each MZI circuit has an input grating coupler, a Y-branch splitter, two waveguide arms, a Y-branch combiner and an output grating coupler. The rings and additional building blocks are modelled using their corresponding library components, as described below and in Section IV-C. The waveguide arms use the effective index, group index, dispersion and dispersion slope of the compact model in Table II, taken at the reference frequency of 193.1 THz, and the arm lengths of the fabricated layout. The grating couplers (TE and TM versions) and Y-branches use measured S-parameter models from the SiEPIC EBeam library [9-11], [14]. The waveguide loss is set to 3 dB/cm, a typical value for strip waveguides in this process [4]. The same Y-branch model (ebeam_y_1550) was used for both polarizations; a back-to-back test with TM grating couplers confirmed that its S-parameter data includes the TM mode.

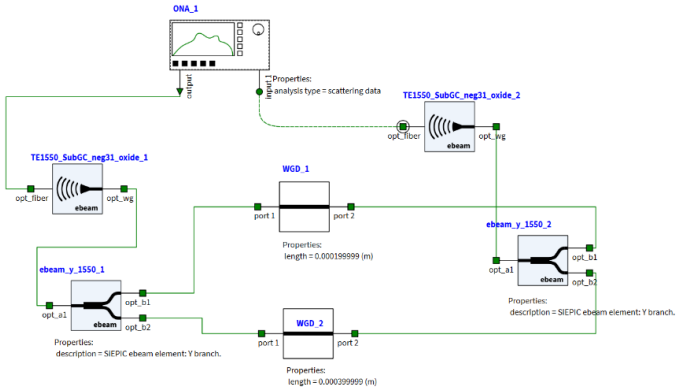
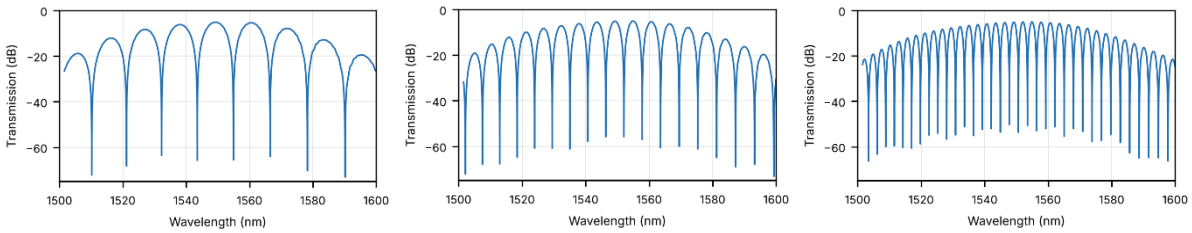


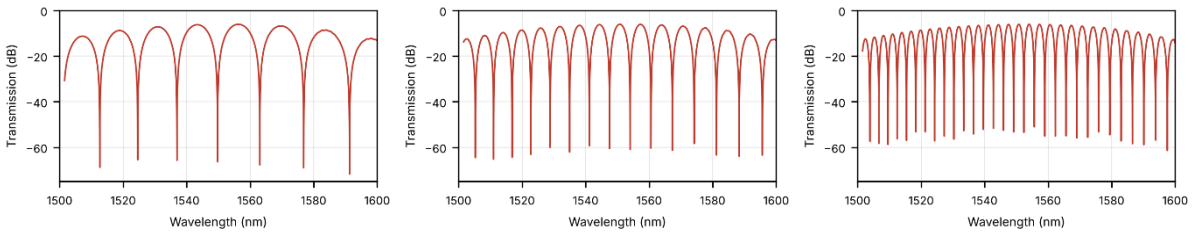
Fig. 6. MZI circuit model in Lumerical INTERCONNECT (TE, shown with the arm lengths of MZI3, 199.999 μm and 399.999 μm). MZI1 and MZI2 use the same circuit with different arm lengths, and the TM circuit uses TM grating couplers.

Figs. 7 and 8 show the simulated spectra. The fringes follow the grating coupler envelope, which peaks near 1550 nm and limits the usable bandwidth. The simulated FSRs near 1550 nm, measured between adjacent minima, are 11.42, 5.71 and 2.86 nm for ΔL = 50, 100 and 200 µm, which agree with the values in Table III to within 0.2%. The FSR increases from 10.9 nm at 1516 nm to 11.9 nm at 1584 nm for MZI1. This increase follows mainly from the λ^2 dependence in (7); the group index changes by only about 0.3% over 1500 to 1600 nm. For the TM interferometers, the simulated FSRs at 1550 nm, interpolated between adjacent minima, are 13.05 nm for MZI4 (ΔL = 49.5 µm), 6.46 nm for ΔL = 100 µm and 3.23 nm for ΔL = 200 µm. MZI4 agrees with Table III, and the two longer devices, which are not on the chip, follow the same $1/\Delta L$ law. The TM FSR changes much more across the band than the TE one: for MZI4 it grows from 11.9 nm near 1519 nm to 14.4 nm near 1584 nm, because the TM group index falls from 4.0 to 3.4 between 1500 and 1600 nm. A TM FSR is therefore meaningful only together with its wavelength.



(a) (b) (c)

Fig. 7. Simulated transmission spectra of the TE MZIs: (a) MZI1, ΔL = 50 µm (FSR = 11.42 nm), (b) MZI2, ΔL = 100 µm (FSR = 5.71 nm), (c) MZI3, ΔL = 200 µm (FSR = 2.86 nm).



(a) (b) (c)

Fig. 8. Simulated transmission spectra of the TM MZIs: (a) MZI4, ΔL = 49.5 µm (FSR = 13.05 nm), (b) ΔL = 100 µm (FSR = 6.46 nm), (c) ΔL = 200 µm (FSR = 3.23 nm). The FSR values are at 1550 nm; the devices in (b) and (c) are not on the chip.

Ring1 was simulated with the directional-coupler half-ring cell from the component library [14] (R = 10 µm, gap 200 nm, L_c = 0), connected to the TE grating couplers and closed with a 36.897 µm waveguide that uses the compact model of Table II. The circuit therefore has the same round-trip length as the layout, 68.3 µm. Fig. 9 shows the circuits and the spectrum. Twelve resonances fall in the 1500 to 1600 nm band, and their spacing grows from 8.0 nm near 1505 nm to 8.9 nm near 1594 nm. At 1550 nm the FSR is 8.45 nm, and (9) gives n_g = 4.16, which is 1.0% below the MODE value and 1% above the 8.37 nm of Table III. The library cell has its own simulated S-parameters for the bend and the coupler, which Table II does not control, so this 1% is the difference between that cell and the MODE-based model. The measurement will show which is closer. A Lorentzian fit of the resonance near 1554 nm (Fig. 9d) gives a full width at half maximum of 0.062 nm, which is a loaded quality factor of about 2.5×10^4. The median width of the interior resonances is 0.06 nm. The extinction ratio changes from resonance to resonance, between about 3 dB and 18 dB. Ring2 (gap 300 nm) could not be simulated with this cell: I set the gap to 0.3 µm and ran it again. The result fields of the cell, which show κ^2 = 0.0216 at 1550 nm for the 200 nm gap, stayed empty, and the

spectrum was identical to that of Ring1. The library therefore has no data for the 300 nm gap, and the Ring1 result is valid for the 200 nm gap only. An estimate for Ring2 is given in Section IV-C. Its weaker coupling should give the same FSR with narrower resonances and a higher Q, and the measurement will test this.

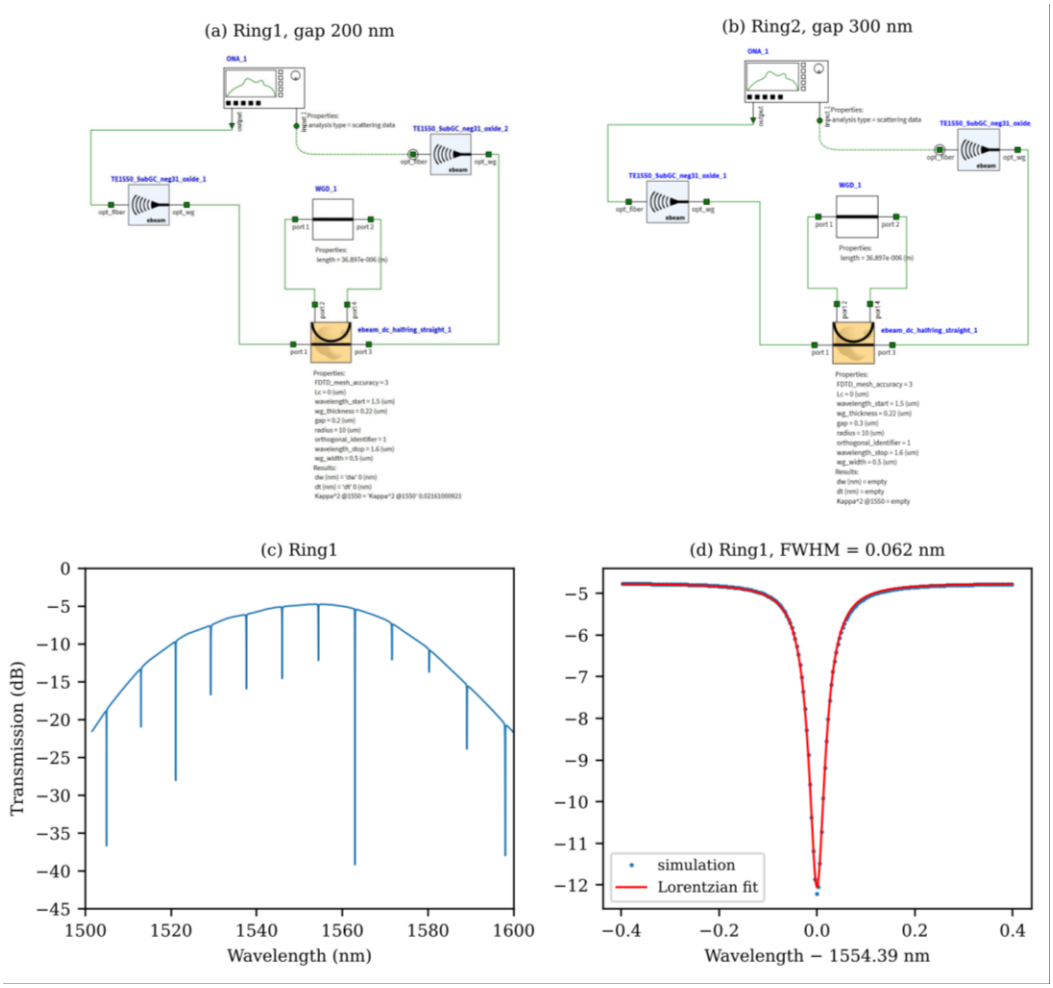


Fig. 9. Rings (R = 10 µm, round-trip length 68.3 µm): INTERCONNECT circuits of (a) Ring1 (gap 200 nm) and (b) Ring2 (gap 300 nm), each with the library half-ring cell closed by a 36.897 µm waveguide; (c) simulated transmission of Ring1 over the full band; (d) resonance of Ring1 near 1554 nm with a Lorentzian fit. The grating coupler response is included. The Ring2 circuit gave the same spectrum as Ring1, so no separate Ring2 spectrum is shown.

G. Planned Analysis of Measured Data

The measured spectra will be processed in four steps:

- Use Loopback1 for the TE circuits; for the TM circuits, which have no TM loopback, fit a low-order polynomial to the smooth envelope and subtract it in dB, without removing the fringes or other features of interest.
- Locate the transmission nulls. Nulls are sharper than maxima, so their positions can be found more precisely.
- Compute the FSR between each pair of adjacent nulls and apply (7) to get n_g at each fringe midpoint, giving $n_g(\lambda)$ across the band.
- As a check, fit the full transfer function (3) to the normalized spectrum using the compact model (10) with n_1, n_2 and n_3 as free parameters, then get n_g from (11).

For the rings, the resonance dips are located the same way, and (9) gives n_g from the spacing between adjacent resonances. Fitting each dip with (8) also gives r and a, which shows how the coupling gap affects the resonance. Because the TE mode has three ΔL values, ng can be extracted three independent ways from the TE MZIs, plus a fourth way from the rings. The single TM device gives one TM value. Agreement between them confirms the analysis. Any consistent offset from the simulated n_g most likely reflects fabricated width or thickness differing from the nominal 500 nm × 220 nm. The corner analysis in Section III-H shows that the TM mode is far more sensitive to thickness than to width, while the TE mode responds to both. Wafer-scale studies of this kind of variation [12] will guide the interpretation.

The other circuits are analysed in the same way, with the loopback spectrum used for the TE circuits to remove the grating coupler response. For Bragg1, the stopband is defined as the wavelength range over which the normalized transmission is at least 3 dB below the reference response. Its measured centre wavelength and width will be compared with the simulated values of 1546.5 nm and 13.5 nm, respectively, and with the fabrication-corner estimates in Section IV-C. For BraggTM1, the measured stopband will be compared with the coupled-mode estimates in Section IV-C and the corresponding fabrication-corner wavelength shifts. For DC1, the normalized transmission of the through port gives the splitting ratio $1 - \kappa^2$, to be compared with the simulated value of 0.78 at 1550 nm and its wavelength dependence.

H. Manufacturing Variability: Corner Analysis

The nominal results above assume a perfect 500 nm × 220 nm waveguide. Real devices vary: the silicon thickness changes across the wafer, and the width changes with lithography, etch and the position on the chip [12]. Both variations affect the modal effective index and group index, influencing the MZI fringe spacing and ring-resonance spacing as well as their absolute wavelengths. Bragg-grating wavelength shifts are estimated separately in Section IV-C. I did a corner analysis to put numbers on this for each waveguide type. The corners combine a width of 470 or 510 nm with a thickness of 215.3 or 223.1 nm, around the nominal 500 nm × 220 nm. They are labelled C1 (470 nm, 215.3 nm), C2 (470 nm, 223.1 nm), C3 (510 nm, 215.3 nm) and C4 (510 nm, 223.1 nm).

Each corner was solved with the same Lumerical MODE model as in Section III-B: the silicon width and thickness were changed, and the mode was tracked from 1500 to 1600 nm in 21 points (100 × 100 mesh cells in the 2.5 μm × 1.7 μm solver window), for the quasi-TE and the quasi-TM mode. The n_{eff} curve of each corner was fitted with the compact model (10), with a largest residual below 4×10^{-5}, and n_g follows from (11); it agrees with the group index reported by MODE to within 0.002. The nominal geometry gives $n_g = 4.204$ (TE) and 3.720 (TM), the values of Table I. The variation of the corners is reported relative to this nominal.

TABLE IV. EFFECTIVE AND GROUP INDEX AT 1550 NM FOR THE PROCESS CORNERS

Corner	Width (nm)	Thickness (nm)	n_{eff} TE	n_g TE	n_{eff} TM	n_g TM
Nominal	500	220.0	2.4468	4.204	1.7688	3.720
C1	470	215.3	2.3724	4.253	1.7167	3.482
C2	470	223.1	2.4039	4.262	1.7664	3.742
C3	510	215.3	2.4416	4.180	1.7443	3.587
C4	510	223.1	2.4722	4.188	1.7963	3.835
Spread (max − min)			0.0998	0.082	0.0796	0.353

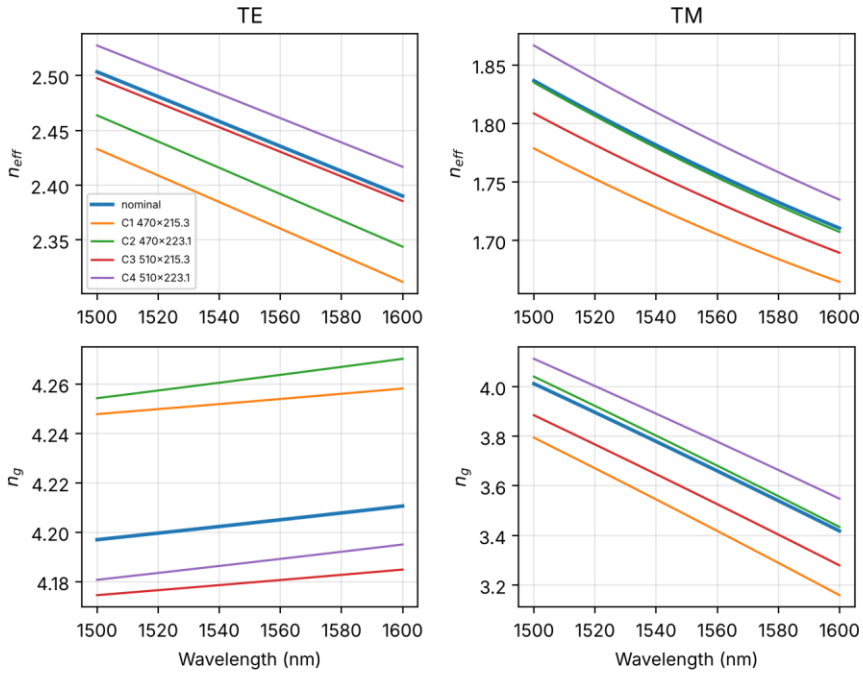


Fig. 10. Effective index (top) and group index (bottom) of the quasi-TE (left) and quasi-TM (right) modes at the five process corners.

Table IV and Fig. 10 show the main trends. For the TE mode, n_{eff} changes by about 1.7×10^{-3} per nm of width and 4.0×10^{-3} per nm of thickness. For the TM mode, the width sensitivity is much smaller (0.7×10^{-3} per nm) and the thickness sensitivity larger (6.5×10^{-3} per nm), because the TM mode is mainly confined by the top and bottom interfaces, so its index is set largely by the 220 nm height. Across the corners, n_{eff} covers 2.372 to 2.472 (TE) and 1.717 to 1.796 (TM). The group index varies much less for TE (4.18 to 4.26, 2.0% of nominal) than for TM (3.48 to 3.83, 9.5% of nominal). This matters for the experiment, because the FSR depends on n_g, not on n_{eff}. In the TE case, a narrower waveguide lowers n_{eff} but raises n_g, as the mode spreads into the cladding and the dispersion grows, so the FSR changes by much less than the peak positions do.

Table V uses these n_g values in (7) and (9) to predict the FSR of each circuit at 1550 nm. The FSR of the TE interferometers changes by only about 2% between the extreme corners, so the group index extracted from them should be close to the nominal value even if the waveguide is not exactly 500 nm × 220 nm. The TM interferometers change by about 10%, mainly through the thickness. A TM group index that differs from Table I by this much therefore does not by itself indicate an error in the simulation or the measurement.

TABLE V. PREDICTED FSR AT 1550 NM FOR THE PROCESS CORNERS

Device	Pol.	ΔL / L (µm)	FSR nominal (nm)	FSR min (nm)	FSR max (nm)	Spread (%)
MZI1	TE	50	11.43	11.27	11.50	1.9
MZI2	TE	100	5.72	5.64	5.75	1.9
MZI3	TE	200	2.86	2.82	2.87	1.9
MZI4	TM	49.5	13.05	12.66	13.94	9.8
Ring1, Ring2	TE	68.3	8.37	8.25	8.41	1.9

The peak wavelength is far less predictable. A change Δn_{eff} moves the fringes by $\Delta\lambda \approx \lambda\Delta n_{eff}/n_g$, and this shift does not depend on ΔL. Table VI lists it relative to the nominal design. The shifts reach −27 nm for TE and −22 nm for TM, which is up to about ten FSRs for the TE device with ΔL = 200 µm (MZI3) and about two FSRs for the 50 µm devices MZI1 and MZI4 (last two columns). Fig. 11 shows this for MZI2 (TE, ΔL = 100 µm) and MZI4 (TM, ΔL = 49.5 µm). The FSR, and hence n_g, is the quantity that stays predictable, which is the reason the extraction method uses spacing between peaks and never their absolute position. The wavelength at which a particular fringe or resonance appears is therefore sensitive to the fabricated geometry and cannot be predicted from the nominal simulation alone. Under a common systematic change in waveguide geometry, the resonance wavelengths of the three TE MZIs are expected to shift in a similar direction. The magnitude of the shift may differ because of local fabrication variations, dispersion and differences in the effective optical paths. Differences between devices should therefore be evaluated against measurement uncertainty and possible device-to-device process variation.

TABLE VI. ESTIMATED FRINGE AND RESONANCE WAVELENGTH SHIFT AT 1550 NM RELATIVE TO THE NOMINAL DESIGN, IN NM AND IN UNITS OF THE NOMINAL FSR

Corner	Shift TE (nm)	Shift TM (nm)	TE shift / nominal FSR (MZI3, ΔL = 200 µm)	TM shift / nominal FSR (MZI4, ΔL = 49.5 µm)
C1	-27.4	-21.7	9.6	1.7
C2	-15.8	-1.0	5.5	0.1
C3	-1.9	-10.2	0.7	0.8
C4	+9.4	+11.4	3.3	0.9

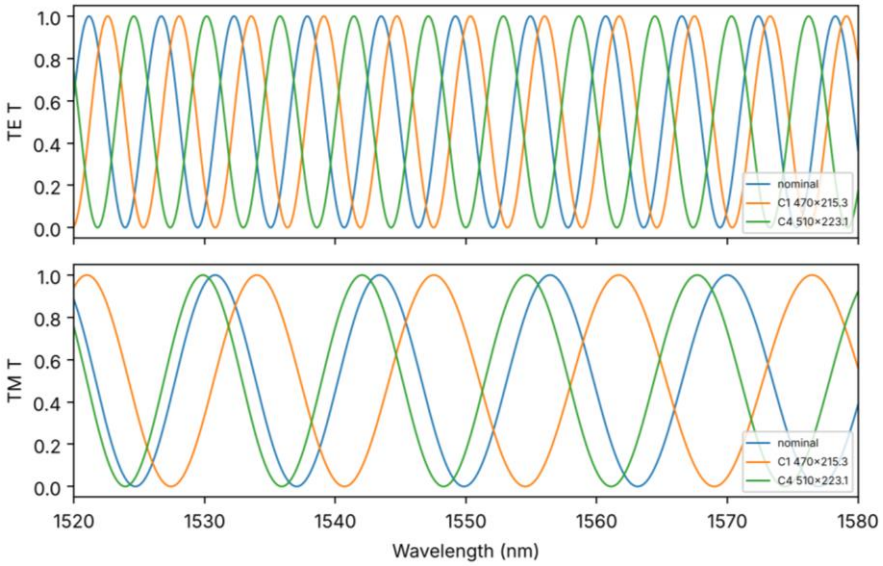


Fig. 11. Transmission of MZI2 (quasi-TE, ΔL = 100 µm, top) and MZI4 (quasi-TM, ΔL = 49.5 µm, bottom) at the nominal geometry and at the C1 and C4 corners. Loss and the grating coupler envelope are not included.

Three limits apply. Only width and thickness were varied, with the four corners taken as fully correlated within a device. Sidewall angle, roughness and the local variation between two arms of the same interferometer were not included. A Monte Carlo analysis would give a statistical spread instead of bounds, which I leave for the comparison with measured data. The coupling gap of the rings also changes with the

process, which moves the coupling coefficient r. This changes the depth and width of the resonances but not their spacing, so Table V is unaffected.

IV. Fabrication

A. Process

The devices will be made in the edX Phot1x fabrication run by electron-beam lithography on a silicon-on-insulator wafer with a 220 nm device layer and a 2 µm buried oxide, and covered by an oxide cladding [5], [11]. The pattern is defined in a single etch step, with a nominal silicon device-layer thickness of 220 nm, as assumed in Section III-A. The nominal width of the straight strip waveguides is 500 nm, while the Bragg gratings use the specified width corrugations listed in Table VII. Fabrication-induced width and thickness variations are represented by the process corners in Section III-H. Devices on the same chip may share systematic process variations, while local variations can occur between devices and between the two arms of an individual interferometer. Table VI therefore illustrates the predicted wavelength shifts under the assumed systematic process corners rather than guaranteeing identical shifts for every device.

B. Layout

The layout was made with KLayout and the SiEPIC-Tools package with the SiEPIC EBeam PDK [14], in the 605 µm × 410 µm area assigned to each student. The grating couplers are placed on a 127 µm pitch so that the input and output fibres of the array can address two couplers at the same time, and each circuit has an optical label (opt_in_TE_1550_device_Farinaaaaaaaz_<name>) that the automated measurement system uses to find it. The TM circuits use the TM grating couplers and larger bend radii. Each circuit was drawn with waveguide paths converted to waveguide components, and the layout was checked with the SiEPIC verification for waveguide, component and label errors until none remained. The final layout (Fig. 12) has ten circuits, listed in Table VII.

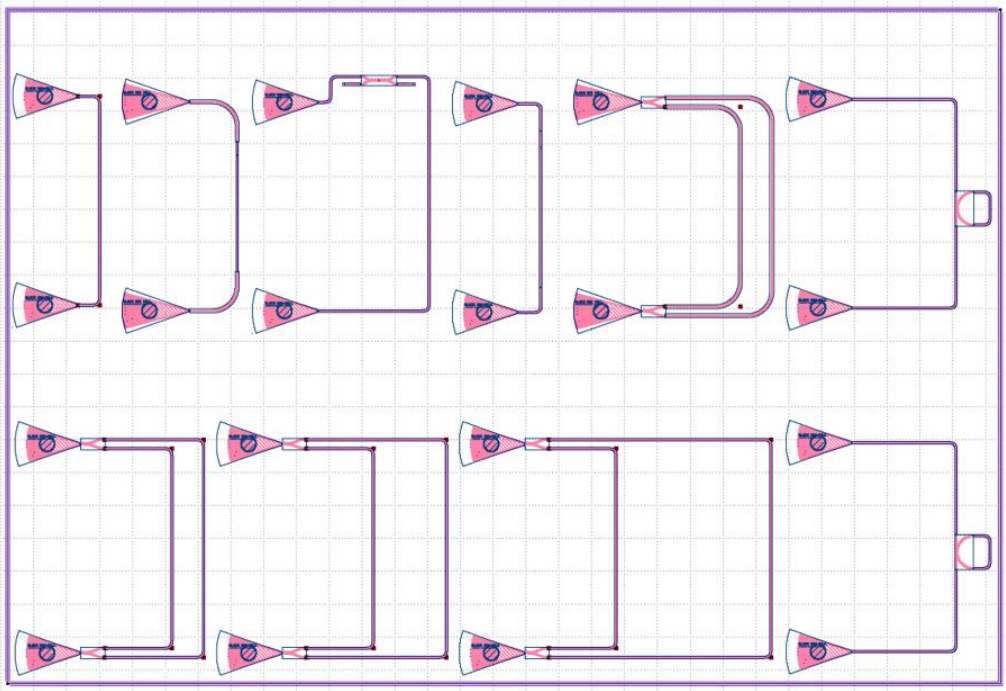


Fig. 12. Final KLayout layout in the 605 µm × 410 µm area. Top row, left to right: Loopback1, BraggTM1, DC1, Bragg1, MZI4 (TM) and Ring2. Bottom row: MZI1, MZI2, MZI3 and Ring1.

TABLE VII. CIRCUITS IN THE SUBMITTED LAYOUT

Cell	Pol.	Description	Key parameters
MZI1	TE	Unbalanced MZI	Arms 199.999 and 249.999 µm, ΔL = 50.000 µm
MZI2	TE	Unbalanced MZI	Arms 199.999 and 300.001 µm, ΔL = 100.002 µm
MZI3	TE	Unbalanced MZI	Arms 199.999 and 399.999 µm, ΔL = 200.000 µm
MZI4	TM	Unbalanced MZI	Arms 200.001 and 249.495 µm, ΔL = 49.494 µm (approx. 49.5 µm)
Ring1	TE	Single-bus ring	R = 10 µm, gap 200 nm, round-trip length 68.3 µm
Ring2	TE	Single-bus ring	R = 10 µm, gap 300 nm, round-trip length 68.3 µm
Bragg1	TE	Bragg grating	300 periods, Λ = 317 nm, corrugation 50 nm, 95.1 µm long
BraggTM1	TM	Bragg grating	180 periods, Λ = 438 nm, corrugation 80 nm, 78.8 µm long
DC1	TE	Directional coupler	L_c = 10 µm, unused ports terminated
Loopback1	TE	Reference waveguide	Two grating couplers, 150.0 µm waveguide

C. Additional Structures

Four structures were added to the interferometers and rings, to cover other building blocks of the platform. Bragg1 is a waveguide Bragg grating for the TE mode, with 300 periods of Λ = 317 nm and a sinusoidal corrugation of 50 nm. The period was chosen using the Bragg condition $\lambda_B = 2\, n_{eff}\, \Lambda$ with n_{eff} obtained from the compact model in Section III-D. This gives a predicted Bragg wavelength of approximately 1551 nm. BraggTM1 is the analogous grating for the TM mode, with 180 periods of Λ = 438 nm and an 80 nm corrugation. The TM mode has a lower effective index (approximately 1.77), so a longer period (438 nm) is required to satisfy the Bragg condition near 1550 nm. The larger corrugation (80 nm) is intended to increase the index modulation and compensate, in part, for the weaker width dependence of the TM mode (Table IV). Both gratings sit in the straight part of a U-shaped waveguide between two grating couplers.

Bragg1 was simulated in INTERCONNECT with the library grating model [14] between the two TE grating couplers, with the number of periods, period and corrugation of the layout. Fig. 13 shows the result. The stopband is centred at 1546.5 nm, 4.5 nm below the 1551 nm of the compact-model estimate. The two calculations use different models of the grating, so the difference is not unexpected, and it is only 0.3% in effective index. The width of the stopband is 13.5 nm, from 1539.8 to 1553.3 nm at 3 dB below the transmission of the grating couplers alone, and the transmission falls by about 50 dB at the centre. The library model reports a Bragg wavelength of 1546.5 nm and a coupling coefficient $\kappa = 6.84 \times 10^4\ m^{-1}$ for this grating. The first-order coupled-mode relation $\Delta\lambda \approx \lambda^2\kappa/(\pi\, n_g)$, with $n_g \approx 4.2$, gives a stopband of about 12 nm, which agrees with the simulated width. Outside the stopband, the grating shows side lobes, the largest at 1535.7 nm and at 1555.2 nm. The Bragg wavelength follows $\lambda_B = 2\, n_{eff}\, \Lambda$. For a fixed grating period, a change in effective index therefore produces a wavelength shift $\Delta\lambda_B \approx 2\Lambda\Delta n_{eff}$. This differs from the MZI fringe shift in Table VI, which is estimated using the group index.

Using the effective-index differences in Table IV gives estimated Bragg-wavelength shifts of approximately −47, −27, −3 and +16 nm for corners C1 to C4, respectively. Applying these shifts to the nominal library-simulated centre wavelength of 1546.5 nm gives estimated corner wavelengths of approximately 1499, 1519, 1543 and 1563 nm. These are first-order estimates referenced to the library result; the compact-model Bragg condition predicts a nominal wavelength of approximately 1551 nm. The

difference between the two nominal predictions should be kept in mind when comparing the corner estimates with measurements. The C1 estimate places the stopband centre at the lower edge of the 1500–1600 nm band, so only part of it could be measured.

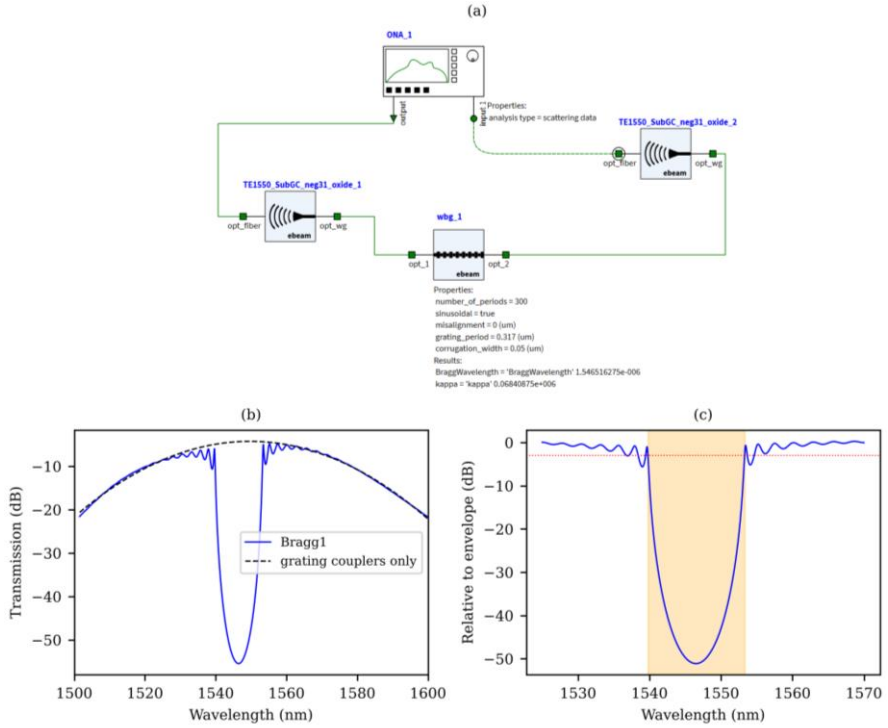


Fig. 13. Simulated Bragg1 (300 periods, Λ = 317 nm, corrugation 50 nm): (a) INTERCONNECT circuit with the library grating model between the two grating couplers; (b) transmission over the full band, with the response of the two grating couplers alone as a dashed line; (c) transmission relative to that response. The shaded region is the 13.5 nm stopband at 3 dB.

There is no TM version of this library model, so BraggTM1 was estimated with the coupled-mode theory of a uniform grating, using Lumerical MODE for the coupling. The effective index was solved at 1550 nm for TM waveguides of 460 and 540 nm width, which are the two widths of the 80 nm corrugation (n_{eff} = 1.7405 and 1.7959), and the coupling coefficient follows from $\kappa = \pi\Delta n/\lambda$ with Δn the amplitude of the index modulation. The same calculation for Bragg1 (475 and 525 nm) gives $\kappa = 8.2 \times 10^4$ m^{-1}, which is 16% above the 6.84×10^4 m^{-1} of the library model, so the TM coefficient was multiplied by 0.84. As a check, the transfer function of a uniform grating with the library κ reproduces the library spectrum of Bragg1: a stopband of 13.4 nm against 13.5 nm, and a minimum of −50.5 dB against −51.1 dB (Fig. 15a). For BraggTM1 the same method gives $\kappa = 4.7 \times 10^4$ m^{-1} and $\kappa L = 3.7$, a stopband of about 11 nm and a rejection of about 26 dB at the centre. A 20% change in κ moves the width between 10 and 13 nm and the rejection between 20 and 33 dB. The nominal TM Bragg wavelength from $\lambda_B = 2n_{eff}\Lambda$ is 1549.5 nm. Applying the approximately 0.3% relative offset observed between the compact-model estimate and the Bragg1 library simulation gives an estimated centre wavelength of about 1545 nm. This correction is an assumption based on the TE grating and has not been independently validated for the TM grating; the predicted centre wavelength therefore has an uncertainty of a few nanometres or potentially more. The estimated TM grating has a lower coupling coefficient than Bragg1 and fewer periods (180 versus 300), so the model predicts lower peak rejection. This prediction depends on the assumed index modulation and requires experimental validation. These are estimates and not a simulation of the grating, and the measurement will test them. The

corner shifts $2\Lambda\Delta n_{eff}$ are −46, −2, −21 and +24 nm for C1 to C4, so BraggTM1 is expected between about 1499 and 1569 nm relative to the 1545 nm estimate.

DC1 is the PDK directional coupler (L_c = 10 µm, 200 nm gap, 500 nm wide waveguides, 5 µm bend radius) with the input and the through port connected to the two grating couplers. Only two grating couplers fit in the 127 µm height of the free area, so the two unused ports are terminated with the PDK terminator. The terminators are placed at least 7 µm from the other waveguides, to avoid reflection and loss in the through path. The coupler was simulated in INTERCONNECT with the library model between the two TE grating couplers, once with the output grating coupler on the through port and once on the cross port (Fig. 14). At 1550 nm the cross-coupled fraction, calculated as the cross power divided by the sum of the two, is 0.22, so 78% of the power stays in the through port. It increases with wavelength, from 0.17 at 1520 nm to 0.24 at 1560 nm and 0.32 at 1600 nm, because the longer wavelength mode extends further into the gap. For a low-loss directional coupler, the normalized through-port power fraction is approximately $T \approx 1 - \kappa^2$, where κ^2 is the cross-coupled power fraction under the same input-power normalization. This relation assumes negligible excess losses and comparable or appropriately corrected through- and cross-port path responses. After normalization using Loopback1, the through-port spectrum can be used to estimate the wavelength-dependent coupling ratio, provided that excess propagation loss and differences between the reference and coupler paths are negligible or appropriately accounted for. Loopback1 is a 150.0 µm waveguide between two grating couplers. Its length is close to the 152 µm path of Bragg1, and it is the reference that gives the response of the TE grating couplers and of the short connecting waveguides, so that it can be removed from the other TE spectra. A TE reference does not apply to the TM devices, because the TM grating coupler has its own response, so the TM measurements use their own fringe or stopband shape and not the absolute transmission.

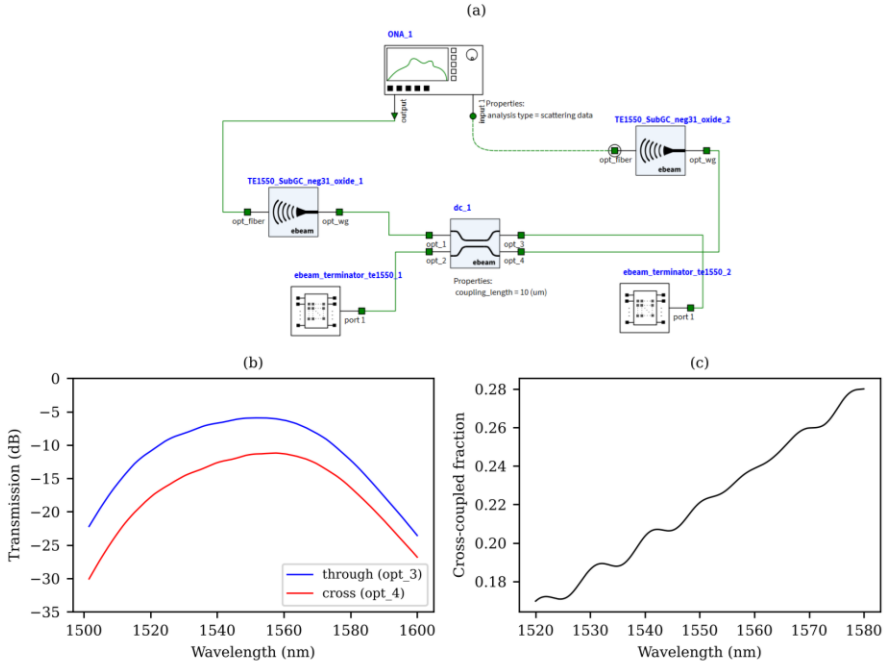


Fig. 14. Simulated DC1 (L_c = 10 µm, gap 200 nm): (a) INTERCONNECT circuit, shown with the output grating coupler on the cross port (opt_4) and the unused ports terminated; for the through-port run it is connected to opt_3; (b) transmission to the through and cross ports, including the grating couplers; (c) cross-coupled fraction.

Ring2 (gap 300 nm) was estimated in a similar way. Lumerical MODE gives the supermodes of two parallel 500 nm waveguides, and their index difference Δn is 0.0204, 0.0131 and 0.0086 for gaps of 200, 250 and 300 nm. The coupling coefficient is proportional to Δn, and the three values follow an exponential decay with the gap, as expected. The power coupling κ^2 of the 300 nm gap is therefore $(0.0086/0.0204)^2 = 0.18$ times that of the 200 nm gap. With the $\kappa^2 = 0.0216$ reported by the library cell for Ring1, this gives $\kappa^2 = 0.0038$ for Ring2. The round-trip amplitude a was taken from Ring1: its resonances are close to critical coupling, so $a \approx r = 0.989$. This reproduces the simulated linewidth of Ring1 (0.058 nm in the model against 0.062 nm in the simulation). With the same a, Ring2 has a linewidth of about 0.034 nm (a loaded Q of about 4.5×10^4) and a resonance depth of about 3 dB, because it is under-coupled (Fig. 15b). The FSR is the same as for Ring1. These Ring2 linewidth, quality-factor and extinction-ratio values are estimates based on the assumed round-trip loss and the coupling coefficient inferred from the MODE supermode calculations. Because the library model did not provide a validated independent simulation for the 300 nm coupling gap, these predictions should not be interpreted as confirmed circuit-simulation results. Measurements of both rings will be used to evaluate the estimates.

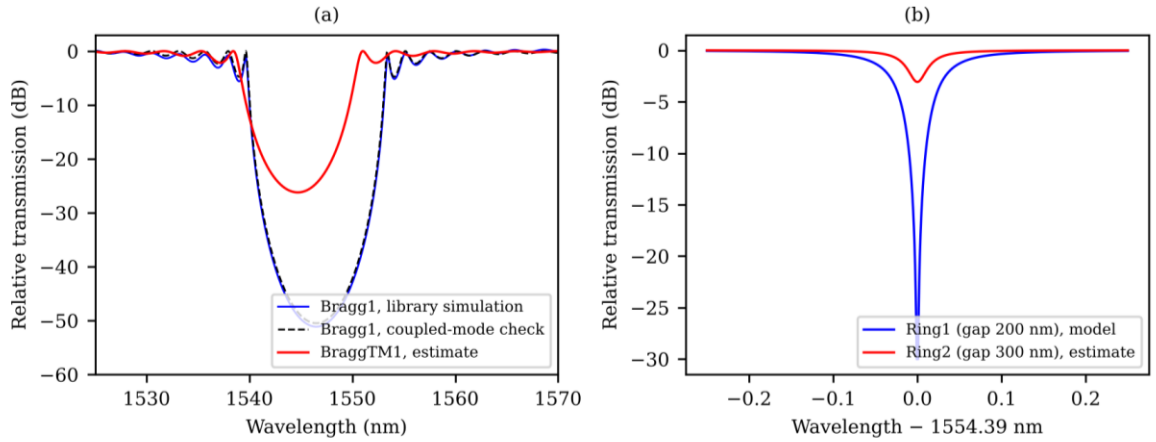


Fig. 15. Estimates obtained with coupled-mode theory and Lumerical MODE: (a) transmission of BraggTM1 relative to the grating coupler response, with the check against the library simulation of Bragg1; (b) the resonance of Ring2 (gap 300 nm) compared with the same model for Ring1 (gap 200 nm).

D. Fabrication Considerations

Section III-H gives the main consequence of the fabrication process for these designs. The fringe positions will be shifted from the simulation by up to a few tens of nm, while the FSR and the extracted n_g should remain close to the simulated values, within the percentages in Table V. For this reason I placed three TE ΔL values, one TM device and two rings on the chip, so that a consistent shift and a consistent n_g can be checked across devices. The Bragg wavelengths of Bragg1 and BraggTM1 move by $2\Lambda\Delta n_{eff}$, which differs between the two polarizations, so they should not shift by the same amount.

V. Experimental Data

Experimental measurements have not yet been completed. The fabricated-device transmission spectra will be presented in this section once the measurement data become available. The analysis will include the MZI fringe positions, ring-resonance wavelengths, Bragg-grating stopbands and directional-coupler transmission.

VI. Analysis

The measured spectra will be analyzed using the procedures described in Section III-G. The extracted group indices will be compared with the MODE predictions and the expected fabrication-corner ranges. The measured ring-resonance linewidths and depths will also be compared with the coupling estimates. Experimental conclusions will be drawn only after the relevant data have been analyzed.

VII. Conclusion

This report presents the design and simulation of silicon photonic test structures for characterizing the quasi-TE and quasi-TM modes of 500 nm × 220 nm strip waveguides. The structures include unbalanced Mach–Zehnder interferometers, TE ring resonators, Bragg gratings, a directional coupler and a reference loopback. The simulations establish nominal group-index predictions and evaluate the sensitivity of the devices to waveguide-width and thickness variations. Experimental validation and comparison with the predicted values remain pending.

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